



Published in final edited form as:

Curr Opin Toxicol. 2022 December ; 32: . doi:10.1016/j.cotox.2022.100373.

Nephrotoxicity of marketed antisense oligonucleotide drugs

Hangyu Wu¹, Aniket Wahane¹, Feryal Alhamadani¹, Kristy Zhang¹, Rajvi Parikh², SooWan Lee¹, Evan M. McCabe³, Theodore P. Rasmussen^{1,3}, Raman Bahal¹, Xiao-bo Zhong¹, José E. Manautou¹

¹Department of Pharmaceutical Sciences, School of Pharmacy, University of Connecticut, Storrs, Connecticut 06269, USA.

²Department of Physiology and Neurobiology, University of Connecticut, Storrs, Connecticut 06269, USA.

³Department of Molecular and Cell Biology, University of Connecticut, Storrs, Connecticut 06269, USA.

Abstract

The field of antisense oligonucleotide (ASO)-based therapies have been making strides in precision medicine due to their potent therapeutic application. Early successes in treating some genetic diseases are now attributed to an emerging class of antisense drugs. After two decades, the US Food and Drug Administration (FDA) has approved a considerable number of ASO drugs, primarily to treat rare diseases with optimal therapeutic outcomes. However, safety is one of the biggest challenges to the therapeutic utility of ASO drugs. Due to patients' and health care practitioners' urgent demands for medicines for untreatable conditions, many ASO drugs have been approved. However, a complete understanding of the mechanisms of adverse drug reactions (ADRs) and toxicities of ASOs still need to be resolved. The range of ADRs is unique to a specific drug, while few ADRs are common to a section of drugs as a whole. Nephrotoxicity is an important concern that needs to be addressed considering the clinical translation of any drug candidates ranging from small molecules to ASO-based drugs. This article encompasses what is known about the nephrotoxicity of ASO drugs, the potential mechanisms of action(s), and recommendations for future investigations on the safety of ASO drugs.

Corresponding author: Dr. Manautou, Jose E. (jose.manautou@uconn.edu), Dr. Zhong, Xiao-bo (xiaobo.zhong@uconn.edu).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Author credit statement

JEM and XBZ conceptualized this review article and produced writing assignments for the trainees that are co-authors. HW, FA, KZ, RP, SL, EMM collected and synthesized all relevant references and collectively wrote the first draft of the article. TPR, RB, ZBZ and JEM revised the manuscript, and HW, AW and JEM edited the final version of the article.

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Keywords

antisense oligonucleotide drug; nephrotoxicity; kidney

Introduction

After two decades of discovery, development, and approvals, antisense oligonucleotide (ASO) drugs have become a new class of nucleic acid therapeutic entities. There is now 14 ASO drugs approved by the US Food and Drug Administration (FDA), with an additional 50 plus in phase I-III clinical trials and more than one hundred in discovery pipelines [1–4]. This novel pharmacotherapeutic approach aims to treat common and rare human diseases [5]. Adverse drug reactions (ADRs) and toxicity are always significant concerns and balancing therapeutic goals with risks of undesired outcomes is part of the intricate process of developing drugs with great clinical value. Eight of the 14 FDA-approved ASO drugs have warnings and precautions for common toxicity, including hypersensitivity reactions, hepatotoxicity, or renal toxicity [6]. Two ASO drugs, Mipomersen and Inotersen, have black warning box labeling due to the risk of hepatotoxicity and nephrotoxicity. Furthermore, due to an unacceptable risk of hepatotoxicity, Mipomersen was withdrawn from the market in 2019 [7], just five years after its approval.

Five ASO drugs approved by the FDA (Nusinersen, Inotersen, Golodirsen, Viltolarsen, and Casimersen) have warning labels for renal toxicity [6]. In the clinical studies of Nusinersen for later onset spinal muscular atrophy (SMA) patients, nearly 70% of Nusinersen-treated patients developed elevated urine protein levels without any elevated serum creatinine or cystatin C levels, suggesting a quantitative spot urine protein test prior to every dose [8]. However, reversible proteinuria has been reported for ASOs with 2'O-Me backbone, highlighting the accumulation of ASOs in kidney as the major reason which directly affects tubular reabsorption of proteins without any tubular damage [9]. In clinical trials of Inotersen [10, 11], approximately 3% of patients developed glomerulonephritis, which may lead to renal failure and the need for dialysis. A high risk of glomerulonephritis has been placed in the black box warning of Inotersen.

Golodirsen, Viltolarsen, and Casimersen are approved to treat Duchenne muscular dystrophy (DMD), a rare genetic disease with various mutations causing exon skipping/inclusion of the dystrophin gene [12]. Although safety assessment of ASO drugs for DMD has been done in pre-clinical studies in various species, due to the small sample sizes in human clinical trials, the risk of renal toxicity for DMD patients still looms. This has led to US FDA warnings for these ASO drugs in humans. Whether or not these ASO drugs cause nephrotoxicity is presently unclear. This review aims to provide an overview of the potential nephrotoxic actions of the ASO drugs based on observations from pre-clinical and clinical studies and what is currently known or hypothesized on the molecular mechanism(s) of toxicant action. This review will begin with a summary of the nephrotoxic effects of conventional small molecule drugs.

Nephrotoxicity caused by conventional small molecule drugs

Proper kidney function is essential to human physiology. As a critical organ of the urogenital system, the kidney maintains organismal homeostasis by regulating osmolarity, maintaining ion concentration and acid-base balance, monitoring pH level, producing essential hormones, and filtering and excreting metabolic by-products and waste [13–15]. The functional unit of the kidney is the nephron. This tiny, yet intricate structure is composed of the glomerulus and Bowman's capsule, the leading site of filtration, a tubular system consisting of the proximal and distal tubules and the loop of Henle, where selective reabsorption occurs, and the collecting duct system, which functions to collect, store, and transport waste to the ureters for bodily excretion. Nephrons produce urine through three successive phases: filtration, reabsorption, and secretion. Filtration, carried out by the semi-permeable capillaries lining the glomerulus, removes proteins and other cellular elements from the prospective waste. Reabsorption, carried out by passive and active transport processes across membranes of the tubular system, transfers water, solutes, and other essential biomolecules, such as glucose and amino acids, back to the circulating blood. The tubular fluid that has been filtered and reabsorbed is then concentrated in the loop of Henle and sent to the collecting duct system, where it is eventually excreted as urine. Agents causing nephrotoxicity, including various pharmaceuticals, are known to impair or disrupt the vital functions of filtration and osmotic homeostasis performed by nephrons.

Drug-induced kidney injury (DIKI) caused by several small-molecule pharmaceuticals is well known. Two prominent examples of DIKI are those caused by cisplatin (a widely used chemotherapeutic agent) and statins (a collection of cholesterol-lowering drugs that target the enzyme HMG-CoA Reductase). In the case cisplatin, renal injury can be induced by various mechanisms. Kidney damage due to cisplatin is associated with its bioaccumulation within the kidney to toxic levels [16]. Insufficient transporter function in the kidney is likely involved in the accumulation of cisplatin and the risk of DIKI [17]. Recent findings have shed light upon the molecular features of cisplatin-induced nephrotoxicity. A recent report showed that cisplatin could induce elevated levels of reactive oxygen species in proximal tubular epithelial cells leading to renal lesions [18]. Two recent reports suggest that cisplatin can cause epigenetic changes in the kidney with potential toxicological consequences [19, 20].

In contrast to the direct renal toxicity caused by cisplatin, statins can lead to kidney damage by indirect mechanisms. Myalgia (muscle pain) is a common side-effect of statins [21, 22]. When statin-induced damage to muscle tissue is extreme, rhabdomyolysis can occur (though infrequently), which releases large amounts of muscle-derived protein degradation products into the bloodstream, which in turn accumulate and cause damage to nephrons [23–25]. These two examples, and others not reviewed here, show that classic small-molecule drugs can induce kidney damage through direct or indirect mechanisms.

Nephrotoxicity caused by some FDA-approved ASO drugs

In contrast to the known mechanisms of renal toxicity caused by certain conventional small molecule drugs, a deeper understanding of the risk of DIKI from ASO drugs is required.

Companies focusing on the development of ASO drugs have highlighted their findings regarding clinical kidney toxicities which sheds some insights on possible mechanisms. The tissue distribution of ASO drugs majorly depends on the route of administration. Intravenous infusion (Golodirsén, Viltolarsén, and Casimersén) leads to higher bioavailability and rapid distribution to organs such as liver, kidney, and spleen. Administration of ASOs by different routes such as subcutaneous injection (Inotersén) or intrathecal injection (Nusinersén) have showcased improved distribution half-life and CNS delivery respectively. The mechanism of ASO metabolism is primarily via nucleolytic breakdown by endo- and exonucleases at the tissue site wherein they are accumulated. Kidney associated toxicities of ASOs are often a result of parent ASO distribution in the kidney along with accumulation of ASO degradation products from different organs such as liver [3]. Certain class of ASOs such as locked nucleic acids (LNA) have reported renal tubule toxicity which doesn't translate to clinic [26]. On the other hand, for siRNAs with N-acetylgalactosamine (GalNAc) conjugation have depicted reduced nephrotoxic risk due to active liver targeting [26]. Direct and/or indirect mechanism(s) by which the ASO drugs can induce kidney toxicity need to be investigated in depth. However, in the following sections, we will review and synthesize the current information on the risk of renal toxicity for each FDA-approved ASO drug.

Nusinersén (Spinraza)

Nusinersén, brand name Spinraza, was developed and manufactured by Biogen, Inc. and was approved by the FDA on December 23rd, 2016, with a New Drug Application number (NDA 209531) [27]. Nusinersén is administered intrathecally. It is the first ASO that directly targets survival motor neuron-2 (SMN2) mRNA splicing for the treatment of spinal muscular atrophy (SMA) in pediatric and adult patients [8]. SMA is caused by a mutation in the *SMN1* gene that stops the production of SMN protein. The disease phenotype is the result of a lack of SMN function and an insufficient number of SMN proteins, leading to muscle weakness, respiratory distress, and paralysis [28]. A nearly identical gene to *SMN1*, namely *SMN2*, produces only 10% functional mRNA, which is insufficient to compensate for the loss of SMN1 due to the exclusion of exon 7 from 90% of the mature SMN2 mRNA transcripts [28]. Interestingly, correction of the *SMN2* exon 7 has been demonstrated therapeutically beneficial to patients with SMN. The mechanism of such exclusion is through the action intronic splicing silencer N1 (ISS-N1) [28, 29]. The ASO drug, Nusinersén, through inhibition of ISS-N1, blocks its ability to exclude exon 7, producing more full-length mRNA, therefore more functional SMN protein is made [28, 29].

The preclinical data on Nusinersén submitted to the FDA (Center for Drug Evaluation and Research, application number: 209531Orig1s000) provide the main toxicological findings highlighted here [30]. In a 13-week subcutaneous injection toxicity study in juvenile CD-1 mice treated with Nusinersén, the kidneys were found to contain small round basophilic granules in both the medium and high dose groups, indicative of oligonucleotide presence in the proximal convoluted tubules in the renal cortex. Since no morphological changes in renal tubules were detected, such as vacuolation, these observations were not considered to be of great toxicological significance. This report states that it remains uncertain whether these observations are incidental findings or somehow associated with the accumulation of basophilic granules. Few studies indicated that the vacuolation in the kidney occurs

because when Nusinersen is taken up by the endosomal/lysosomal system, it is eluted during tissue fixation and processing, leaving dilated vacuolar system remnants. Therefore, the relationship of this observation as a causative factor in renal toxicity is still uncertain [30]. However, renal tubular vacuolation is actually noted with other ASOs.

In a one-year repeated dose toxicity study in juvenile cynomolgus monkeys (*Macaca fascicularis*) with a 26-week recovery phase from Nusinersen treatment, the toxicokinetic studies demonstrated that drug concentrations in kidneys were commonly higher than in the CNS. Liver concentrations of Nusinersen were commonly lower than in all other organs examined [30].

The current clinical data on the nephrotoxicity of Nusinersen remains inconclusive and controversial. According to the FDA drug labeling, 17 of 51 (33%) patients treated with Nusinersen had elevated urine protein, compared to 5 of 25 (20%) in control patients [8]. In patients with later-onset SMA with a mean treatment duration of 34 months, 36 of 52 (69%) showed elevated urine protein [8]. However, there were no elevations in serum creatinine to cystatin C ratios. By contrast, a shift from high blood urea nitrogen to creatinine has been noted in a few cases [31]. Furthermore, in these cases, proteinuria did not occur more frequently in Nusinersen-treated patients than in healthy people [31]. Therefore, these studies did not prove renal impairment by Nusinersen treatment [31–34]. The most frequent laboratory abnormality in SMA patients is proteinuria [35]. However, the proteinuria ceased and was commonly attributed to sampling contamination [35]. Another study evaluated Nusinersen's safety in 16 adult patients with SMA clinical sub-types 3 and 4. Patients with type 3 SMA, also known as Kugelbert-Welander, develop symptoms between 18 months and early adulthood. This form of SMA worsens over time but is not significantly life-threatening. Type 4 SMA is scarce and usually does not appear until beyond the age of 30. All patients treated with Nusinersen for a period of over 14 months, regardless of SMA type, showed no kidney dysfunction [36]. No significant increases in either serum creatinine or urea nitrogen were detected either. However, using creatinine as a marker of kidney function is of little utility in SMA patients because its homeostasis is significantly affected in multiple neuromuscular diseases [31].

There are two upcoming clinical trials for Nusinersen. One is an observational clinical trial ([NCT04419233](#)), recruiting 50 participants to evaluate the safety of Nusinersen in the post-marketing setting in China, with an estimated completion date of January 2023. Another observational clinical trial ([NCT04317794](#)) is expected to recruit 145 participants to evaluate its safety in South Korean patients. The data from these studies will offer more information regarding the potential renal toxicity of Nusinersen.

Inotersen (Tegsedi)

Inotersen, known commercially as Tegsedi, is manufactured by Ionis Pharmaceuticals and was approved for use by the FDA on October 5th, 2018 for the treatment of the polyneuropathy of hereditary transthyretin (TTR)-mediated amyloidosis (hATTR-PN) in adults [37]. This is a rare disease, with an estimated 10,000 afflicted patients worldwide [38]. This is an inherited, progressive, and fatal disease. Current treatment options include liver transplantation and the TTR tetramer stabilizers Tafamidis and Diflunisal hypothesized

to prevent TTR from undergoing conformational changes resulting in amyloid fibrils formation [39]. Recently, more promising gene-silencing therapies have been developed, including Patisiran, the first FDA-approved siRNA drug [40], and the ASO drug Inotersen. This latter causes the degradation of mutant and wild-type TTR mRNAs through binding to the TTR mRNAs, which reduces serum TTR protein and TTR protein deposits in tissues [41].

One of Inotersen's reported ADRs is glomerulonephritis. However, pharmacokinetics and toxicology preclinical studies in mice and monkeys showed that Inotersen is well-tolerated without noticeable adverse effects [42]. Furthermore, a phase 1 clinical trial consisting of 65 healthy volunteers who received Inotersen at a dose of 400 mg by subcutaneous injection (SC) for four weeks, showed excellent tolerability and safety, without significant adverse effects noted, including kidney function abnormalities [42]. An international, randomized, double-blind, placebo-controlled, phase 3 trial NEURO-TTR ([NCT01737398](#)) was conducted to assess the efficacy and safety of Inotersen. Patients were randomly assigned in a 2:1 ratio to receive weekly SC injections of Inotersen (300 mg) or placebo. Glomerulonephritis occurred in 3% of patients in the Inotersen group compared with no patients in the placebo group [11]. The open-label extension (OLE) of the NEURO-TTR study was conducted to assess the efficacy and safety of extended treatment with Inotersen for up to 5 years ([NCT02175004](#)) [43]. Among 135 NEURO-TTR patients enrolled in the OLE, 85 received Inotersen, and 50 switched from placebo to Inotersen. There were no cases of acute glomerulonephritis in the OLE.

In those human cases where glomerulonephritis is observed, the proposed mechanism is the accumulation of ASOs in proximal tubule cells, which leads to increased tubular proteinuria. Urine protein to creatinine ratio (UPCR) more significant than five times the upper limit of normal (ULN) occurred in 15% of Inotersen-treated patients, compared to 8% of patients on placebo. Therefore, Inotersen should generally not be initiated in patients with a UPCR of 1,000 mg/g or greater [11].

Golodirsen (Vyondys 53)

Golodirsen was approved by the FDA on December 12th, 2019 and is manufactured by Sarepta Therapeutics. It is used to treat DMD, an inherited disorder with progressive muscular weakness caused by mutations in the dystrophin gene. Golodirsen works by inducing the skipping of exon 53 in the dystrophin pre-mRNA [44].

Renal toxicity was observed in preclinical studies, specifically in male mice, rats, and monkeys, documented in the Non-Clinical Review 211970Orig1s000 for Golodirsen by the Center for Drug Evaluation and Research of FDA [45]. Ultrastructural anomalies in the urinary bladder have also been observed in male mice. Two studies conducted in male mice, a 12-week study using an intravenous (IV) injection and a 26-week study via SC injection, provided evidence of Golodirsen-induced renal adverse events. In the 12-week study, microscopic images showed tubular dilatation, basophilic or eosinophilic casts, and vacuolation in kidneys, which correlated with increases in serum markers of renal function (e.g., urea, nitrogen, and creatinine). These effects were primarily observed at the highest dose tested. Furthermore, transitional hypertrophy epithelium of both the ureter and urinary

bladder was observed at all doses. In the 26-week study, degeneration of the transitional epithelium of the urinary bladder and renal tubular degeneration was observed at all doses. IV administered Golodirsen was also tested in male rats. This was a 13-week study that resulted in tubular degeneration at all, except the lowest doses tested. At the highest amount, the microscopic changes detected were accompanied by increased serum urea nitrogen. In a 39-week male monkey study, IV administration of Golodirsen resulted in various adverse kidney outcomes, including basophilia, dilatation, or mononuclear cell infiltration at all doses, which correlated well with increases in serum markers of renal function (urea, nitrogen, creatinine), principally at the highest dose tested.

For the first-in-human multicenter phase 1 and 2 clinical trials of Golodirsen, kidney toxicity was not observed in the 23 participants of this study [46]. However, as a precaution due to the significant risk of renal toxicity observed in animal studies, FDA warnings and precautions for renal toxicity have been issued for Golodirsen in its drug labeling [47]. An upcoming post-marketing clinical trial is currently conducted by Sarepta Therapeutics. This is a double-blind, placebo-controlled phase 3 trial named ESSENCE ([NCT02500381](#)) with an estimated enrollment of 222 participants to evaluate the efficacy and safety of Golodirsen to support its accelerated approval. The trial results are expected to be released in April 2024.

There is a potential risk for clinical renal toxicity with Golodirsen because of the strong evidence derived from nonclinical studies showing irreversible, sometimes fatal, renal damage in juvenile animals with only a small margin of safety. Since creatinine may not be a reliable biomarker of renal injury in DMD patients because of their reduced skeletal muscle mass, renal function in this patient population should be monitored using other standard parameters beyond creatinine clearance. Therefore, a 24-hour measurement of glomerular filtration rate (GFR) before initiation of therapy is recommended [47]. Furthermore, monitoring proteinuria by dipstick urinalysis and monitoring of serum cystatin C every three months during Golodirsen treatment is also recommended. If the dipstick reading shows proteinuria of 2+ or greater, or serum cystatin C is significantly elevated, then a 24-hour urine collection to quantify proteinuria and assess GFR should be performed [47].

Viltolarsen (Viltepso)

Viltolarsen, manufactured by Nippon Shinyaku Co, and marketed under the tradename Viltepso, was granted accelerated approval by the FDA on July 12th, 2020, for treating patients diagnosed with DMD [48]. Viltolarsen is specifically designed to target patients who carry mutations in exon 53 of the dystrophin gene. Binding Viltolarsen to the dystrophin mRNA results in exon 53 skipping during mRNA processing [49]. This exon exclusion leads to increased dystrophin production, which helps keep muscle cells intact.

Kidney toxicity of Viltolarsen was observed in preclinical studies, documented in the Non-Clinical Review 212154Orig1s000 for Viltolarsen by the Center for Drug Evaluation and Research of FDA [50]. Viltolarsen was administered to juvenile male mice via SC injection on postnatal day seven, then once weekly via IV injection from postnatal days 14 to 70. It was administered at 0, 15, 60, 240, and 1,200 mg/kg, with renal toxicity observed at the highest doses. Mice in the 240 and 1,200 mg/kg dose groups showed a dose-dependent

increase in mortality and greater incidence and severity of renal tubular effects, such as degeneration. Reduced body weight and delayed sexual maturation were also observed at the highest dose. No renal toxicity was observed at 60 mg/kg, and plasma exposures were similar to that in humans at a dose of 80 mg/kg.

Although renal toxicity was not observed in clinical trials with a limited patient sample size, including clinical trial [NCT02740972](#) [51] and clinical studies in Japanese population [52, 53], the FDA still warns of the potential risk of developing renal toxicity and fatal glomerulonephritis with Viltolarsen in the drug labeling [54]. It is suggested that serum cystatin C, urine dipstick, and UPCR should be measured before starting Viltolarsen treatment. Urine dipstick results should be monitored monthly, and serum cystatin C and urine protein-to-creatinine ratios should be examined every three months. If persistent increases in serum cystatin C or proteinuria are detected during treatment, kidney toxicity may occur, which warrants further monitoring of kidney function.

Casimersen (Amondys 45)

Casimersen, brand name Amondys 45, was developed by Sarepta Therapeutics and approved on February 25th, 2021, with the NDA number 213026 [55]. It is a new ASO of the phosphorodiamidate morpholino oligomer subclass with a primary use for the treatment of DMD patients who have confirmed mutations in exon 45 of the dystrophin gene [56].

Kidney toxicity of Casimersen was evaluated in preclinical studies, documented in the Non-Clinical Review 213026Orig1s000 for Casimersen by the Center for Drug Evaluation and Research of FDA [57]. In the preclinical studies, general toxicity analysis was conducted in male mice for 12, 22, and 26 weeks of treatment and in male cynomolgus monkeys with treatment durations of 12 and 39 weeks. The most significant organ toxicity was abnormal kidney function. In mouse studies, increases in plasma urea nitrogen and creatinine were observed at moderate and high doses of Casimersen (960 and 2,000 mg/kg). The clinical pathology effects were accompanied by increased regeneration/degeneration of kidney tubular epithelium, which was not reversed during the recovery period. In the male cynomolgus monkey studies, histopathology showed effects in the kidneys as cytoplasmic basophilia of the tubular epithelium, and tubular vacuolation in all tested Casimersen doses up to 640 mg/kg. The severity and incidence were dose related. Vacuolated macrophage aggregation was observed in mesenteric lymph nodes. Renal tubular basophilia and vacuolation were also observed in high-dose animals in the recovery period. The cytoplasmic basophilia and vacuolation of renal tubular epithelium are common microscopic findings observed after treatment with ASO drugs [58]. Administration of Casimersen to juvenile male rats via IV injection also led to renal tubular degeneration/necrosis at the highest dose tested.

The safety and tolerability of Casimersen were examined in a phase 1/2 clinical study ([NCT02530905](#)) [59]. In phase 1 or 2 study, 12 patients were all male and aged 7–21 years, diagnosed with DMD and confirmed genetic mutations amenable to exon 45 skipping. This was a randomized, double-blind, dose-titration, placebo-controlled study. The patients were randomized 2:1 to receive escalating doses of Casimersen or placebo once a week through IV infusion for 12 weeks [59]. This study found no evidence of a significant

kidney abnormality or toxicity risk. A phase 3 clinical trial ESSENCE ([NCT02500381](#)) with Casimersen in patients with DMD is recruiting participants with an estimated enrollment of 222 [60]. This is a double-blind, placebo-controlled, multi-center study to evaluate the efficacy and safety of Casimersen. The estimated study completion date is April 30, 2024. Another phase 3 extension study to evaluate Casimersen or Golodirsén in patients with DMD ([NCT03532542](#)) is also recruiting with an estimated enrollment of 260 participants [61]. This is a long-term, non-randomized, open-label extension study with an estimated completion date of August 10, 2026. Completion of these phase 3 clinical trials with larger participant sizes will provide further evidence of whether Casimersen can result in kidney toxicity or not in DMD patients.

Although kidney toxicity was not observed in the phase 1/2 clinical study with limited participants (12 for both treatment and control) [59], the FDA still includes warnings and precautions for kidney toxicity in Casimersen's drug labeling [62]. Like Golodirsén and Viltolarsén, serum cystatin C, urine dipstick, and protein-to-creatinine ratio should be measured before starting Casimersen. GFR should also be measured before starting the drug. If a persistent increase in serum cystatin C or proteinuria is detected, Casimersen should not be prescribed. According to the FDA label, this risk warning is not in response to signs of kidney toxicity, such as glomerulonephritis, in the clinical trials, but instead, it is reflective of the risk for ASO class of medications as a whole [58].

Considering the very recent approval of Casimersen by the FDA and the limited number of patients for which this drug is approved, post-marketing safety surveillance data is essentially non-existent. More data and information must be gathered to better estimate its risk for renal toxicity and overall safety profile.

Possible molecular mechanisms of ASO drug-induced nephrotoxicity

Figure 1 provides a graphical depiction that summarizes what is known and/or proposed to be the mechanism(s) of nephrotoxic actions of ASO drugs. Although the data on renal toxicity of ASO drugs are insufficient, ASO drugs mainly concentrate in the liver and kidney during absorption, distribution, metabolism, and excretion [3]. Therefore, the need for additional clinical studies is critical to understand the nephrotoxicity mechanisms of ASO drugs fully. According to the skeleton and chemical structure of this class of drugs, they can be divided into phosphorothioate (PS) ASOs, and phosphorodiamidate morpholino oligomers (PMO). For PMOs, the sugar is replaced by the morpholino group, and the phosphate linkages are replaced by phosphorodiamidate ones [63]. For PS ASOs, the phosphate linkages are replaced by phosphorothioate ones (PS). 2' modification to the sugar-phosphate backbone of PS ASOs is the most common one, and among all the substitutions, the most useful one is 2'-MOE [64]. PMOs include Golodirsén, Viltolarsén, and Casimersén, while PS 2'-MOE ASOs have Nusinersén and Inotersén. Inflammation and apoptosis are proposed as possible mechanisms for the adverse effects of PS ASOs, including nephrotoxicity [5]. Figure 2 summarizes the different chemistries and chemical modifications present in ASO drugs. Table 1. highlights the chemistry, properties, mechanism of action and target mRNA of the reviewed FDA approved ASO drugs.

Some in vitro as well as animal and human studies have documented that PS ASOs can activate the alternate complement system, resulting in immunological and cytotoxic effects [65]. Interference of PS ASOs with the complement cascade via inhibition of Factor H, can lead to multi-organ infiltration of inflammatory cells and in the worst-case scenario, mortality from circulatory collapse [65]. Factor H inhibition by ASOs has been particularly problematic in non-human primates' studies, which is the most sensitive animal species for this effect [65, 66]. On the contrary, humans are much less susceptible to complement activation by ASOs with no evidence of this type of sequelae. Again, given the limited number of patients with rare diseases treated with ASOs, the risk and magnitude of these potential adverse events, including nephrotoxicity, remain uncertain.

Some ASO drugs have been reported to induce direct cell death [67]. Recently, three molecular mechanisms have been described that apply to toxicities of ASO drugs to various organ systems. The first is the primary molecular mechanism of PS ASO toxicity. PS ASOs with greater risks of producing adverse effects have the tendency to bind to proteins more tightly than ASO drugs with a better safety profile. The avid interactions between an ASO drug with certain proteins containing RNA recognition motifs leads to RNase H1-dependent paraspeckle protein compartmentalization into nucleolus-like structures. Paraspeckles are small and irregularly sized subnuclear bodies containing RNA-protein structures and found in the interchromatin space of mammalian cells [68]. These ASO drugs generated structures can affect pre-rRNA synthesis, leading to nucleolar stress and apoptotic cell death [69–71]. The second proposed mechanism is that certain ASOs may hybridize with off-target RNAs of critical genes, resulting in degradation of the RNA of the off-target gene and loss of its protein product. If such target gene is essential for cell viability, cell death ensues [72]. Nevertheless, 2'-MOE modified PS ASOs have exhibited a higher clinical safety and tolerability, including low adverse reaction observations from both human clinical trials and non-human primate studies [66, 73, 74]. Despite this, the clinical safety of 2'-MOE ASO drugs, as regards to nephrotoxicity still concerns the US FDA.

It is speculated that the above mechanism of inflammation and apoptosis may be closely related to the degeneration of the transitional epithelium of the urinary bladder, deterioration of renal tubule, and irreversible renal damage in preclinical studies of Golodirsen; the decline of renal tubule in preclinical studies of Nusinersen; the degeneration of renal tubule observed at the highest doses in animal experiments receiving Viltolarsen; the deterioration of renal tubule in preclinical studies of Casimersen; and the glomerulonephritis and proteinuria in clinical trials of Inotersen. For locked nucleic acids (LNA) which offer higher binding affinities to target sequences, elevation of extracellular growth factor (EGF) was identified as in vitro biomarker which was able to predict LNA induced cytotoxicity in human kidney tubular cells [75]. These EGF in vitro screens further identified the role N-acetylgalactosamine (GalNAc) for attenuating cytotoxicity of LNA as compared to their unconjugated counterparts [76]. Such in vitro safety assessment studies are beneficial for the development of newer ASO candidates. In the published literature, clinical trial evaluation data for these emerging drugs are still very scarce, and the mechanism of renal toxicity is very little explored. Therefore, future scientific research can develop in this direction.

Conclusion

This review discussed the existing nephrotoxicity data of five approved ASO drugs in preclinical studies and clinical trials. Their effects on the kidney mainly include renal tubular degeneration, glomerulonephritis, and increased urinary protein levels. However, definite experimental conclusions of the nephrotoxicities of ASO drugs are insufficient, even after their approval for use as prescription drugs. Although renal toxicity caused by ASOs does not appear to be a generalized observation in clinical studies with small number of patients, further research is necessary, when considering that the strength of the data comes from preclinical studies. Although the toxicological mechanisms and risks for nephrotoxic events of ASO drugs remain unclear this article presents the potential toxicological mechanism(s) according to the existing information that also provides the benchmark for follow-up work.

Conventional drugs commonly target and modulate the function of proteins, producing sound and efficacious curative effects. However, for many diseases, proteins cannot be effectively targeted for various reasons. With a unique targeting strategy, small nucleic acid drugs modify the upstream mRNA molecules that produce proteins. This innovative precision medicine approach can solve many congenital gene defects for which curative solutions cannot be presently achieved. ASOs have excellent market potential. Currently, the ASO drugs developed mainly aim to treat a limited number of rare genetic diseases. However, the disease resolution potential of this novel therapeutic approach for a wide array of chronic conditions is immeasurable. One of the biggest obstacles in the development and successful clinical use of ASO drugs is ensuring their efficacy and safety. Our understanding of safety concerns about ASO drugs is in the nascent stages, but there are already emerging concerns about toxicities associated with the novel class of medications. In 2019, Mipomersen was withdrawn from the market due to the high risk of hepatotoxicity [7]. Furthermore, nephrotoxicity is a concern with the use of several ASO drugs. However, and as stated previously, the data in support of kidney dysfunction comes mostly from pre-clinical studies. Up to this date, there is no preponderance of evidence of adverse events in kidneys and renal functions from clinical studies and post-marketing surveillance. What hinders our ability to truly understand the potential risk for nephrotoxicity by ASO drugs in humans is the limited number of patients treated with these drugs. With the development and subsequent approval of ASO drugs for treating diseases with high prevalence rates, a better understanding of the risk of this class of drugs will be attained.

In conclusion, precision medicine is the future of health care, but the benefit to risk ratio for patients also needs to be considered with respect to better drug safety. As ASO drugs accumulate in kidneys after metabolism, it is necessary to conduct carefully designed research to understand why these degradation products accumulate in kidneys and the sequelae of cellular events in nephrotoxicity. This new knowledge will be critical in the chemical design of the next-generation ASO drugs with optimal renal safety.

Funding

This study was partly supported by the National Institutes of Health (NIH) National Institute of General Medical Sciences [R35GM140862 to X.B.Z.], the National Cancer Institute [R01CA241194 to R.B.] and the University of Connecticut Research Excellence Program [JEM].

ABBREVIATIONS:

ADRs	adverse drug reactions
ASO	antisense oligonucleotide
DIKI	drug-induced kidney injury
DMD	Duchenne muscular dystrophy
NDA	New Drug Application
OLE	open-label extension
PMO	phosphorodiamidate morpholino oligomers
PS	phosphorothioate
US FDA	US Food and Drug Administration
GFR	glomerular filtration rate
IV	intravenous
SMN	survival motor neuron
SC	subcutaneous injection
SMA	spinal muscular atrophy
TTR	transthyretin
ULN	upper limit of normal
UPCR	urine protein to creatinine ratio

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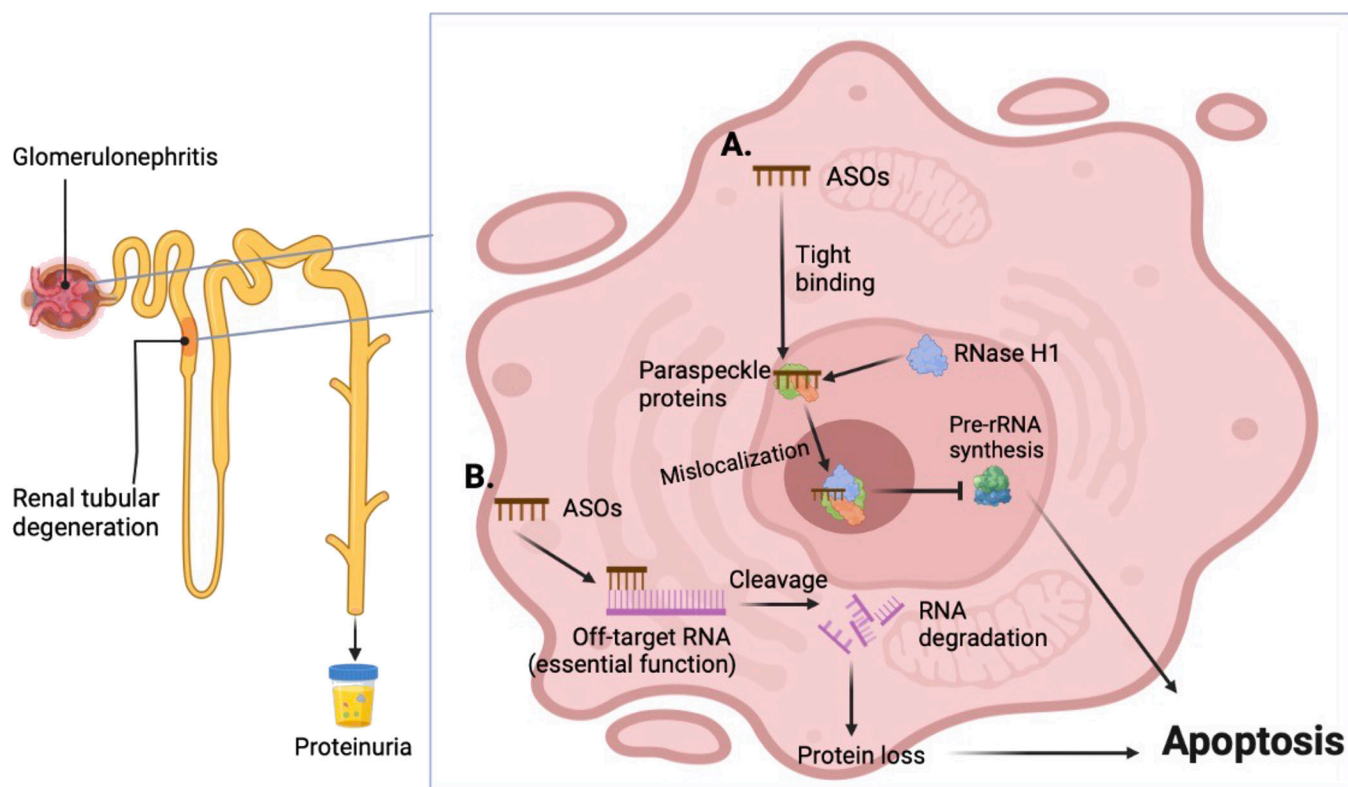


Figure 1.

Proposed molecular mechanisms of ASO drug-induced nephrotoxicity. The main nephrotoxic findings with ASO therapeutics consists of glomerulonephritis, renal tubular degeneration, and proteinuria. There are three potential molecular mechanisms that may explain these findings: (A). After entering the nucleus, some ASO drugs bind tightly to paraspeckle proteins in an RNAase H1-dependent manner. This complexation triggers its localization to the nucleolus, where it interferes with pre-ribosomal RNA synthesis, inducing apoptosis; (B). ASO drugs can also bind to untargeted RNAs of genes essential for cell viability, causing their degradation and subsequent loss of the encoded proteins that are essential for cell survival; and (C). Certain ASO drugs with specific chemical features, such as 2' F containing molecules, may similarly bind to paraspeckle proteins, but instead of mislocalization of the complex to a nucleosomal compartment, ASO binding alters the conformation and function of the adducted protein, which can produce apoptotic cell death.

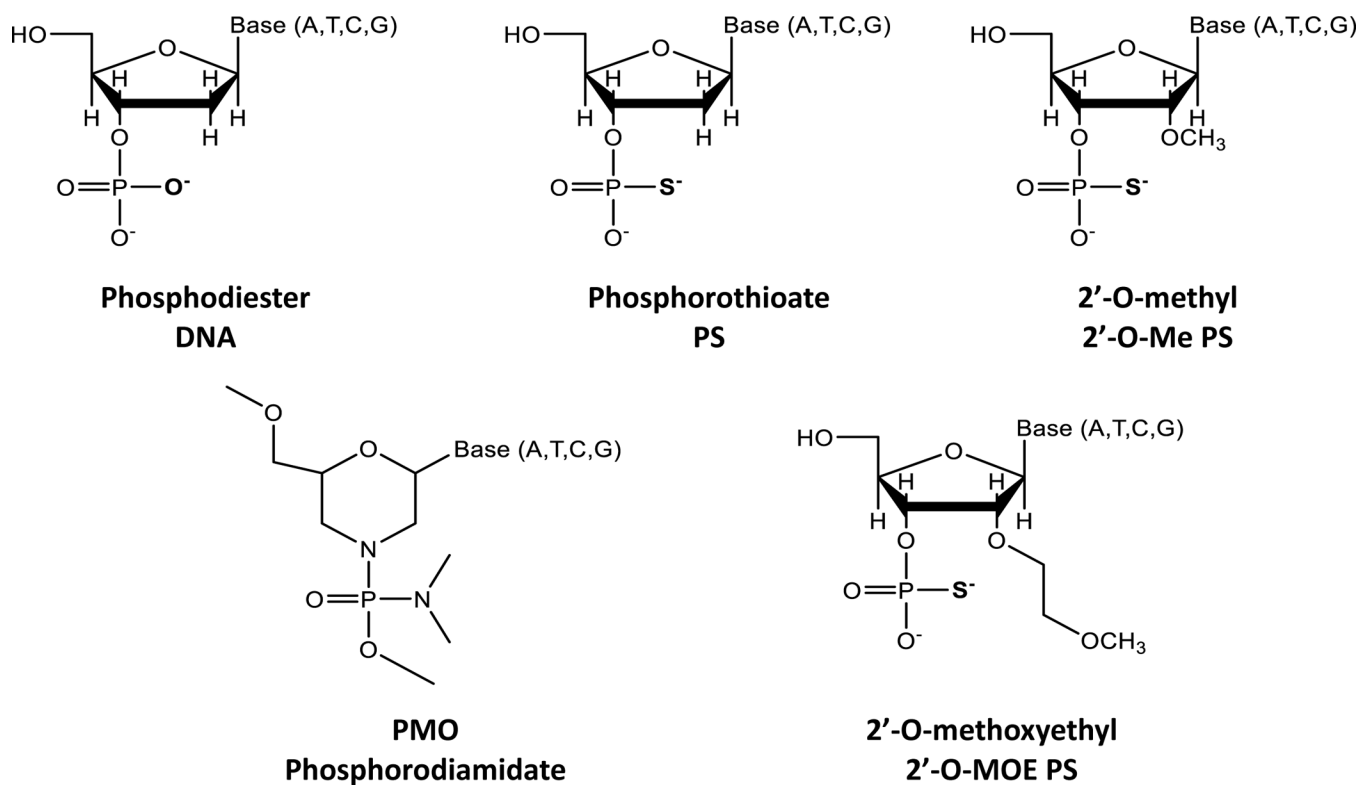


Figure 2:
Different chemistries and chemical modifications of ASO drugs.

Table 1:

List of reviewed US FDA approved drugs.

ASO name	Chemistry	Properties	Mechanism of action	Target mRNA
Nusinersen	2'-O-MOE	Higher binding affinity, enzymatic stability, reduced immune stimulation	RNase H activation & cleavage	ISS-NI
Inotersen	2'-O-MOE	Higher binding affinity, enzymatic stability, reduced immune stimulation	RNase H activation & cleavage	TTR
Golodirsen	PMO	Better aqueous solubility, higher binding affinity	Steric hindrance	Dystrophin
Viltolarsen	PMO	Better aqueous solubility, higher binding affinity	Steric hindrance	Dystrophin
Casimersen	PMO	Better aqueous solubility, higher binding affinity	Steric hindrance	Dystrophin

Abbreviations: 2'-O-MOE, 2'-O-methoxyethyl; PMO, Phosphorodiamidate; ISS-NI, intronic splicing silencer N1; TTR, transthyretin.